Counting more women's work: unpaid helpers, subsidiary activity, and the rise in women's employment in India, 2017–18 to 2023–24

Abstract

India's Periodic Labour Force Survey (PLFS), the official national labor force survey, shows the worker-population ratio of women aged 15 and over rising from 22.0 percent in 2017–18 to 40.3 percent in 2023–24, a near doubling under unchanged definitions of work. Using unit-level data from the two nationally representative survey rounds (163,000 and 160,042 women aged 15 and over), I reproduce the published estimates and decompose the rise with exact additive accounting, design-based standard errors, and a Kitagawa shift-share decomposition. The new finding is where the rise lies. Own-account work and unpaid help in household enterprises account for 86 percent of the 18.4-point rise, agriculture for 73 percent, and women counted as workers only through a subsidiary activity alongside domestic duties for 48 percent. Changes in population composition by state, sector, age, education and marital status do not account for the rise. Weekly hours worked per woman rose by 40 percent, against 84 percent for the worker-population ratio. In 2023–24, 42 percent of self-employed women in primary production produced only or mainly for household consumption; excluding such work, as the 2013 international definition of employment does, would put the ratio at 31.5 to 37.0 percent. The rise is concentrated in the forms of work that are most sensitive to how surveys probe. For India, and for other countries adopting the 2013 standards, headline employment rates are best read alongside principal-status, hours-based, and own-use-adjusted measures.

Keywords: women's employment; measurement of work; unpaid family workers; own-use production; labor force surveys; India

Introduction

Between 2017–18 and 2023–24, India's official labor force survey recorded a near doubling of women's employment. The worker-population ratio (WPR) of women aged 15 and over in usual status rose from 22.0 to 40.3 percent, and the published series rose in every year in between (National Sample Survey Office, 2024). The definitions of work did not change. This paper shows where, in the classification of work, the additional workers are found. Own-account work and unpaid help in household enterprises account for 86 percent of the 18.4-point rise; women

whose main activity remained domestic duties and who were counted as workers only through a subsidiary economic activity account for 48 percent; and agriculture accounts for 73 percent. Mean weekly hours of work per woman rose by 40 percent, about half the 84-percent rise in the WPR. And in 2023–24, 42 percent of self-employed women in primary production produced only or mainly for the household's own consumption, work that the 2013 international standard classifies as own-use production rather than employment (International Labour Organization, 2013).

These results matter beyond India for two reasons. First, many countries are revising their labor force surveys to implement the 2013 resolution of the International Conference of Labour Statisticians (ICLS), which separates employment (work for pay or profit) from own-use production work. Survey experiments in Tanzania, Ghana and the countries of the International Labour Organization (ILO) pilot program show that measured women's employment depends on how questions are asked and how carefully enumerators probe for agricultural and household-based activity (Bardasi et al., 2011; Benes and Walsh, 2018; Gaddis et al., 2021). Those studies vary the questionnaire within a single survey. India's recent experience is a national-scale case of the same sensitivity outside an experiment: a headline rate that moved by 18.4 points under formally unchanged definitions, with the change concentrated in exactly the categories that the experimental literature identifies as most sensitive to probing. Second, women's employment rates are widely used as indicators of progress and as targets in national and international policy. If a large part of a measured rise is subsidiary and own-use work, the headline rate alone can overstate changes in women's paid work, earnings and economic independence, and the measures needed to interpret it have to be published alongside it.

The rise also bears on population policy, because women's employment is treated as a demographic variable as well as an economic one. Projections of the labor force, assessments of a demographic dividend, and discussions of fertility, age at marriage, schooling and the care of children and older persons all use women's employment as an input or as a pathway. These uses assume that a higher recorded rate means a change in how women allocate time between market work, household production and care. If the recorded rise lies mainly in work that sits beside domestic duties, then extrapolating the rate into future labor supply, or reading it as evidence of a changed bargaining position within households, may overstate what has changed. The earlier

literature on the decline in participation linked women's work to household income, education and social norms (Jayachandran, 2021; Klasen and Pieters, 2015). The links between employment and those outcomes are unlikely to be the same for every form of work. A rate that mixes wage work, unpaid family help and subsidiary activity cannot support a single demographic reading. This is a reason to publish the components of the rate along with the rate, and it is a point about interpretation, not a criticism of the survey.

A rise of this size has prompted a debate about its meaning. Studies using published tables and unit-level data show that it is concentrated among rural women and in self-employment, and that earnings from self-employment did not keep pace (Ara and Shrivastav, 2025; Azim Premji University, 2023; Thakur and Chaudhary, 2025). One interpretation is that it reflects distress, with women entering low-productivity family work as household incomes came under pressure (Azim Premji University, 2023; Jha, Ghatak, and Singh, 2026). Another is that additional family labor in agriculture is productive and therefore a positive development (Goldar and Aggarwal, 2026). A third is that part of the rise reflects how completely women's existing activity is recorded as work. Thakur and Chaudhary (2025) argue that revised instructions in the 2023–24 survey contributed to moving women from outside the labor force into self-employment. Women's work in rural India is often part-time, seasonal, home-based and close to the line between production for the market, production for the household's own use and domestic work, and surveys are known to be sensitive to how such work is probed (Bardasi et al., 2011; Beneria, 1981; Deshpande and Kabeer, 2024; Hirway and Jose, 2011; Langsten and Salen, 2008).

This paper does not try to choose between these interpretations directly. Instead, it provides the accounting that any interpretation needs, and it adds four elements that the recent studies do not combine. First, it reproduces 59 of 60 published headline estimates from the unit-level data and decomposes the change in the women's WPR exactly, with design-based standard errors, by status in employment, by industry and, which turns out to be central, by whether women work in a principal or only in a subsidiary capacity. Second, it measures the rise in hours of work per woman, a measure that does not depend on where the line between workers and non-workers is drawn. Third, it uses a question introduced in the 2023–24 survey on the destination of output to estimate a women's WPR consistent with the 2013 international definition of employment. Fourth, it dates the changes in the published field instructions against the annual path of the

published series. That comparison qualifies the instruction-based explanation: about half of the rise had occurred before the worked examples that the survey's own note dates to 2022–23. A shift-share decomposition then tests whether changes in population composition account for the rise.

Five findings stand out. First, own-account work and unpaid help in household enterprises account for 86 percent of the rise, and regular wage employment for 10 percent. Second, 48 percent of the rise consists of women whose principal activity is outside the labor force, typically domestic duties, and who are counted as workers because of a subsidiary economic activity of at least 30 days in the year; 73 percent of the rise is in agriculture. Third, the rise is much smaller when measured in hours: the mean weekly hours worked per woman in the population rose by 40 percent, against 84 percent for the WPR, and the share of women workers who worked fewer than 20 hours in the reference week rose from 20.2 to 32.0 percent. Fourth, in 2023–24, 42 percent of self-employed women in primary production reported that their output was only or mainly for the household's own consumption; excluding such work, as the 2013 international standard would, lowers the 2023–24 WPR of women from 40.3 to between 31.5 and 37.0 percent. Fifth, changes in age, education, marital status, sector and state composition do not explain the rise; it occurred within these groups, and it was largest among rural women aged 35 to 59, women with little schooling and Scheduled Tribe women.

I also document that the field instructions for the 2023–24 survey contain an explicit list prompting enumerators to probe for work in kitchen gardens, orchards, household poultry and dairy and free collection, and worked examples on when such activity makes a woman engaged in domestic duties a subsidiary worker (National Sample Survey Office, 2023a). These were not in the 2017–18 instructions (National Sample Survey Office, 2016), and the survey's own note dates the worked examples to clarifications first provided for the 2022–23 survey (National Sample Survey Office, 2023b). The definitions of work did not change, and the data cannot show how much the instructions contributed. The point is narrower: the categories in which the rise is concentrated are exactly those in which responses depend most on probing, so the rise should be read with that in mind.

The paper contributes to three literatures. For research on women's employment in India, which has largely studied the earlier decline (Afridi, Dinkelman, and Mahajan, 2018; Klasen and

Pieters, 2015; Lahoti and Swaminathan, 2016; Mehrotra and Parida, 2017; Sarkar, Sahoo, and Klasen, 2019), it provides a design-based accounting of the recent reversal that separates principal from subsidiary work and persons from hours. For the literature on the measurement of women's work (Bardasi et al., 2011; Beneria, 1981; Benes and Walsh, 2018; Gaddis et al., 2021; Langsten and Salen, 2008), it adds a large non-experimental case in which a large change in the measured rate is concentrated in the most measurement-sensitive categories. And for the implementation of the 2013 ICLS standards, it shows how a single destination-of-output question allows a survey that keeps the older, broader concept of work to report an employment rate consistent with the new definition. The analysis is descriptive. It does not attribute the change to any particular cause or policy.

Background

Women's work in India and its measurement

For more than a decade, research on women's employment in India was about decline. Successive rounds of the National Sample Survey recorded falling participation of women in the labor force between 2004–05 and 2011–12, despite rapid economic growth, falling fertility and rising education (Afridi, Dinkelman, and Mahajan, 2018; Andres et al., 2017; Chatterjee, Desai, and Vanneman, 2018; Desai and Joshi, 2019; Klasen and Pieters, 2015; Mehrotra and Parida, 2017). Explanations emphasized rising household incomes and an income effect on women's labor supply, rising enrollment in education, the shortage of jobs considered suitable for women, social norms that restrict women's mobility, and changes in the structure of agricultural work (Afridi, Dinkelman, and Mahajan, 2018; Chatterjee, Desai, and Vanneman, 2018; Jayachandran, 2021; Klasen and Pieters, 2015; Mehrotra and Parida, 2017). The cross-country literature on a U-shaped relation between development and women's participation also predicts that women withdraw from family farm work as household incomes rise and return to the labor force later, through wage employment in services (Gaddis and Klasen, 2014; Goldin, 1994; Klasen, 2019). A rise in women's employment that comes mainly through self-employment in agriculture runs against both of these patterns, which is one reason the recent rise has been questioned (Ara and Shrivastav, 2025; Jha, Ghatak, and Singh, 2026).

Measurement has been a concern throughout. Women's work in family farms and enterprises is often not reported as work unless respondents are asked about it specifically, and the answers depend on the wording of questions, on the reference period and on who answers (Bardasi et al., 2011; Beneria, 1981; Langsten and Salen, 2008). In India, time-use data have long recorded more economic activity by women than labor force surveys (Hirway and Jose, 2011), and the distinction between work for income, expenditure-saving work for the household and unpaid domestic work is important for understanding women's work (Deshpande and Kabeer, 2024). Earlier panel evidence also shows frequent movements of Indian women into and out of work (Sarkar, Sahoo, and Klasen, 2019), so the measured rate depends on how intermittent work is captured.

Recent evidence on the rise

A recent literature examines the post-2017 rise directly. Ara and Shrivastav (2025) use the published PLFS tables and unit-level data to show that the rise is driven by rural women and by own-account and unpaid work in agriculture and the unorganized sector, while the share of women in regular salaried jobs fell. Azim Premji University (2023) report that women's employment rates rose after 2019 through an increase in self-employment and that real earnings from self-employment declined over the same period, and read the rise as distress-led. Jha, Ghatak, and Singh (2026) focus on the growth of unpaid family workers and argue that it reflects distress-driven employment growth. Goldar and Aggarwal (2026) question the premise that unpaid female family labor in Indian agriculture has very low productivity, which supports a more positive reading. Thakur and Chaudhary (2025) use unit-level PLFS data and argue that revised instructions and guidelines in the 2023–24 survey contributed to shifting a sizable number of women from outside the labor force into self-employment, alongside distress-driven diversification indicated by falling real incomes in the activities that absorbed women's labor.

These studies agree that the rise is rural and in self-employment, and they disagree about what it means. None of them, to my knowledge, separates the rise into principal and subsidiary capacity with design-based standard errors, compares the rise in persons with the rise in hours per woman, uses the destination-of-output question to estimate an employment rate consistent with the 2013 international definition, or tests composition with a shift-share decomposition. These are the elements that distinguish more work from more workers and work for the market from own-use

production. The dating of the instruction changes also matters for the measurement explanation: if the clarifications came late in the period, they cannot account for the part of the rise that occurred before them.

Measuring women's work beyond India

The measurement problem is general. In agricultural economies, a large part of women's work is on family farms, in livestock and in collection, often for the household's own use and often combined with domestic work. Plot-level data from six African countries show that women's share of crop labor is substantial and varies with how labor is recorded (Palacios-Lopez, Christiaensen, and Kilic, 2017). A survey experiment in Tanzania found that women's measured employment depended on whether a short or detailed questionnaire was used and on whether the woman answered for herself (Bardasi et al., 2011). A survey experiment in Ghana found that farm labor recorded with different recall and respondent designs differed substantially (Gaddis et al., 2021). The ILO's pilot studies for the 2013 standards tested alternative questionnaires in several countries and recommended keeping specific recovery questions on agricultural activity where subsistence or mixed agricultural work is common (Benes and Walsh, 2018).

The 2013 standards change what is counted as employment. Production of goods mainly for the household's own final use, such as subsistence farming, kitchen gardens, household poultry and collection of firewood and water, is a form of work but not employment (International Labour Organization, 2013). Countries that adopt the standards therefore face two sources of change in the measured employment of women at once: a narrower definition, which lowers it, and more careful probing of agricultural and household activities, which raises recorded work. Distinguishing the two requires exactly the information that the PLFS provides in 2023–24: the capacity in which work is done, hours, and the destination of output. India's experience shows what a headline rate can do when the second source operates without the first.

The conceptual change in the 2013 standards is easy to misread. The resolution does not say that production for the household's own use is not work. It says that such production is work outside the category of employment, and it asks countries to measure employment and own-use production work separately (International Labour Organization, 2013). The purpose is to keep an employment indicator that is tied to market participation and income, while making the unpaid

productive work of households, which falls disproportionately to women, visible in its own statistics. The PLFS follows the older production boundary of the national accounts, so it counts own-use primary production as economic activity (National Sample Survey Office, 2016, 2023a). This is a different concept, not an error. The destination-of-output question brings the two concepts together in one instrument, but it classifies work by purpose and not by activity. The answer depends on the respondent's view of what the household mostly does with its output. That mix can vary across seasons and years, and a household may sell in some periods and consume in others. For this reason the lower and upper bounds should be read as a range that depends on the classification rule, and not as a precise estimate of employment in the international sense. The pilot work that informed the standards stressed that recovering agricultural and subsistence activity requires specific questions (Benes and Walsh, 2018), which is the same sensitivity that makes the classification hard.

Concepts in the PLFS

The PLFS classifies each person by usual activity status over the 365 days preceding the survey. The principal status is the activity on which a person spent a relatively long time. A person whose principal status is outside the labor force, such as domestic duties, is counted as a worker in a subsidiary capacity if she was engaged in an economic activity for at least 30 days in the year. The headline 'usual status (ps+ss)' measure counts both principal and subsidiary workers. Status in employment distinguishes own-account workers and employers, unpaid helpers in household enterprises, regular wage or salaried employees and casual laborers. A separate current weekly status classifies persons by their activity in the seven days before the survey, and the survey records the hours actually worked on each of those days.

India's national definition of economic activity follows the System of National Accounts production boundary. It includes all production of primary goods, including free collection of uncultivated crops, firewood and fish, whether for sale or for the household's own consumption (National Sample Survey Office, 2016, 2023a). Two codes for persons outside the labor force are relevant: code 92 (attended domestic duties only) and code 93 (attended domestic duties and was also engaged in free collection of goods, sewing, tailoring or weaving for household use). A woman who does domestic work and also collects firewood or vegetables for the household is

therefore recorded either as code 93 or, if the collection amounts to 30 days or more in the year, as a subsidiary own-account worker.

The PLFS continues to use the broader concept of work that preceded the 2013 resolution. In 2023–24, however, the survey asked self-employed persons in primary-sector activities whether their output was for own consumption only, mainly for own consumption, mainly for sale or entirely for sale. This question was not asked in 2017–18.

Changes in the field instructions

The definitions above are the same in the field instructions for both survey years. The 2023–24 instructions add three elements that the 2017–18 instructions do not contain (Table A1). First, the list of examples of economic activities ‘for which special care is needed’, which enumerators are told to probe so that such activities are ‘not omitted’, has a new item: ‘Working in kitchen garden, orchards, household poultry, dairy, free collection of fish, small game, wild fruits, vegetables, etc. for household consumption/sale’ (National Sample Survey Office, 2023a). Second, a new box of worked examples explains how to distinguish codes 92 and 93 from subsidiary self-employment. In one example, a woman whose main activity is domestic duties and who collects firewood for the household for an hour or two a day for at least 30 days in the year is to be recorded with principal status 92 and a subsidiary economic activity as an own-account worker, not as code 93. Third, a further box applies the same logic to the daily activity record of the current weekly status. The survey’s note on updated instructions states that the clarifications in the two boxes of worked examples were first provided for the 2022–23 survey; it does not date the new item in the list (National Sample Survey Office, 2023b). They clarify rather than change the concepts, but they direct attention to the activities that are most likely to be under-reported among women.

Data and Methods

Data

I use the unit-level data of the first visit of PLFS 2017–18 and PLFS 2023–24, released by the National Statistics Office of the Ministry of Statistics and Programme Implementation. The first-visit schedules cover the rural and urban samples and are the basis of the official annual

estimates in usual status. The files contain 433,339 persons in 102,113 households and 12,773 first-stage units in 2017–18, and 418,159 persons in 101,920 households and 12,743 first-stage units in 2023–24. The analysis sample is persons aged 15 and over: 163,000 women in 2017–18 and 160,042 women in 2023–24, with men used for comparison. The data are anonymized official survey records released for research use; the study involved no contact with participants, and no ethics approval was required for this secondary analysis.

Estimates use the survey multipliers as prescribed in the data documentation: the multiplier is divided by 100 when the sub-sample and the sub-round match and by 200 otherwise, and by the number of quarters to give annual estimates. Standard errors account for the stratified multistage design, with first-stage units within quarter as primary sampling units and the survey strata and sub-strata within state and sector as strata, using the survey package in R (Lumley, 2004). Because the two years are independent samples, the standard error of a change is the square root of the sum of the two squared standard errors. I harmonize state codes across years by combining Jammu and Kashmir with Ladakh and Dadra and Nagar Haveli with Daman and Diu, so that both years refer to the same territories.

Definitions

A person is a usual-status worker if her principal status is a work code (11–51), or if her principal status is not a work code and she reports a subsidiary economic activity with a work code. Workers are classified by the status of the principal activity for principal workers and of the subsidiary activity for subsidiary-only workers, into own-account workers and employers (codes 11–12), unpaid helpers (21), regular wage or salaried employees (31) and casual laborers (41, 51). Agriculture is NIC 2008 divisions 01–03. Weekly hours are the hours actually worked in all economic activities on the seven days before the survey. These are the official definitions, and with them the estimates reproduce the published figures (see Results).

Measures beyond the headline rate

Three measures complement the headline WPR. The principal-status WPR counts only women whose main activity in the year was economic. Mean weekly hours of work per woman in the population, including zeros for women who did not work, is a measure of the volume of work that is not affected by where the line between workers and non-workers is drawn. For 2023–24,

an own-use-adjusted WPR excludes self-employed women in primary-sector activities whose output was for own consumption only, or only or mainly; this approximates the 2013 international definition of employment, as discussed under Results.

Decompositions

Because the WPR is the sum of the shares of the population in each status, the change in the WPR is the exact sum of the changes in the status-specific ratios, and the same holds for the split into principal and subsidiary-only workers and into agriculture and other industries. To assess whether the rise reflects changes in the composition of the population, I use a Kitagawa decomposition (Kitagawa, 1955). With cells defined by state, rural or urban sector, age group (15–24, 25–34, 35–44, 45–59, 60+), education (below primary, primary or middle, secondary or higher secondary, graduate or above) and marital status, the change in the WPR is

where is the population share of cell , its WPR, and bars denote averages of the two years. The first term is the composition component and the second the rate component. Cells present in only one year (0.32 percent of the population) are excluded from the decomposition and the shares are renormalized.

Use of AI tools

The author used Claude (Anthropic) to improve the language and readability of parts of the text and to assist with code review and checking. The author reviewed and verified all outputs and takes full responsibility for the content.

Results

Reproducing the official estimates

The estimates from the unit-level data reproduce the published figures. For 60 published values (the LFPR, WPR and unemployment rate in usual status and the LFPR and WPR in current weekly status, for men and women, in rural areas, urban areas and the whole country, in both years), 59 are identical after rounding to one decimal place and the remaining one (the LFPR in current weekly status for men in 2017–18, published as 75.1 and estimated as 75.16) differs by

0.06 points. Table 1 shows the national values. The women's WPR rose from 22.0 to 40.3 percent, a rise of 18.4 points (95 percent CI 17.7 to 19.0). Men's WPR rose by 5.1 points, from 71.2 to 76.3 percent. The women's unemployment rate fell from 5.6 to 3.2 percent, so almost all of the rise in participation took the form of employment.

Table 1 *Published and reproduced headline estimates, persons aged 15 and over, India (percent)*

Indicator	Sex	2017–18 published	2017–18 estimated	2023–24 published	2023–24 estimated
LFPR, usual status	Women	23.3	23.29	41.7	41.66
LFPR, usual status	Men	75.8	75.80	78.8	78.80
WPR, usual status	Women	22.0	21.98	40.3	40.34
WPR, usual status	Men	71.2	71.17	76.3	76.28
Unemployment rate, usual status	Women	5.6	5.64	3.2	3.15
Unemployment rate, usual status	Men	6.1	6.10	3.2	3.19
LFPR, current weekly status	Women	21.1	21.08	35.6	35.57
LFPR, current weekly status	Men	75.1	75.16	77.5	77.54
WPR, current weekly status	Women	19.2	19.18	33.8	33.81
WPR, current weekly status	Men	68.6	68.65	73.8	73.78

Note. Published values from the NSO press note of 23 September 2024. Estimated values are computed from the unit-level data with the official weights. 59 of 60 published values (including rural and urban values not shown) are reproduced exactly after rounding.

Where the rise occurred

Table 2 and Fig. 1 decompose the change in the women's WPR. Nationally, own-account work and employers added 8.0 points and unpaid helpers 7.8 points, together 86 percent of the rise of 18.4 points. Regular wage employment added 1.8 points and casual labor 0.8 points. Agriculture accounts for 13.4 points, or 73 percent. The rise in principal workers was 9.6 points; the share of women who are workers only in a subsidiary capacity rose almost five-fold, from 2.3 to 11.0 percent, and accounts for 48 percent of the rise.

The rise was concentrated in rural areas. Among rural women the WPR rose by 22.9 points, from 23.7 to 46.5 percent. Unpaid helpers added 10.5 points, of which 6.5 points were subsidiary-only helpers, and own-account workers added 10.1 points, of which 3.5 were subsidiary-only. Subsidiary-only workers account for 50 percent of the rural rise and agriculture for 81 percent. Regular wage employment of rural women rose by 1.2 points, and casual labor in a principal capacity did not change (-0.1 points).

Urban women's WPR rose by a smaller 7.8 points, and the composition was different: regular wage employment rose by 3.4 points, own-account work by 3.1 points and unpaid help by 1.5 points, while casual labor fell slightly. Most of the urban rise was in principal status and outside agriculture.

Men's employment changed in the same direction but on a much smaller scale. Men's WPR rose by 5.1 points; unpaid help rose by 1.9 points and subsidiary-only work by 1.6 points, from 0.5 to 2.1 percent of men. The shift toward self-employment and subsidiary work therefore affected both sexes, but for women it was large enough to nearly double the measured employment rate.

Table 2 *Decomposition of the women's worker-population ratio by status, capacity and industry, women aged 15 and over (percent of population)*

	Rural 2017–18	Rural 2023–24	Rural change (SE)	Urban 2017–18	Urban 2023–24	Urban change (SE)	All India 2017–18	All India 2023–24	All India change (SE)
Worker-population ratio	23.7	46.5	22.9 (0.46)	18.2	26.0	7.8 (0.45)	22.0	40.3	18.4 (0.35)
By status in employment									
Own-account worker or employer	4.5	14.6	10.1 (0.26)	4.3	7.4	3.1 (0.24)	4.4	12.4	8.0 (0.20)
Unpaid helper	9.1	19.6	10.5 (0.34)	2.0	3.5	1.5 (0.16)	7.0	14.8	7.8 (0.24)
Regular wage or salaried	2.5	3.6	1.2 (0.14)	9.5	12.9	3.4 (0.32)	4.6	6.4	1.8 (0.14)
Casual labor	7.5	8.7	1.2 (0.25)	2.4	2.2	-0.2 (0.16)	6.0	6.7	0.8 (0.18)
By capacity									
Principal status	20.9	32.4	11.5 (0.42)	17.1	22.2	5.1 (0.43)	19.7	29.3	9.6 (0.32)
Subsidiary status only	2.8	14.1	11.4 (0.30)	1.0	3.8	2.7 (0.17)	2.3	11.0	8.8 (0.22)
Own-account, subsidiary only	0.7	4.2	3.5 (0.15)	0.5	1.9	1.4 (0.12)	0.7	3.5	2.8 (0.11)
Unpaid helper, subsidiary only	1.5	8.0	6.5 (0.23)	0.3	1.1	0.8 (0.08)	1.1	5.9	4.8 (0.16)
By industry									
Agriculture	17.3	35.8	18.5 (0.43)	1.6	3.2	1.5 (0.18)	12.5	26.0	13.4 (0.31)
Other industries	6.3	10.8	4.4 (0.25)	16.5	22.8	6.3 (0.43)	9.5	14.4	4.9 (0.23)

Note. Usual status (ps+ss). Components in each panel sum to the worker-population ratio. Agriculture is NIC 2008 divisions 01–03. Standard errors account for the survey design.

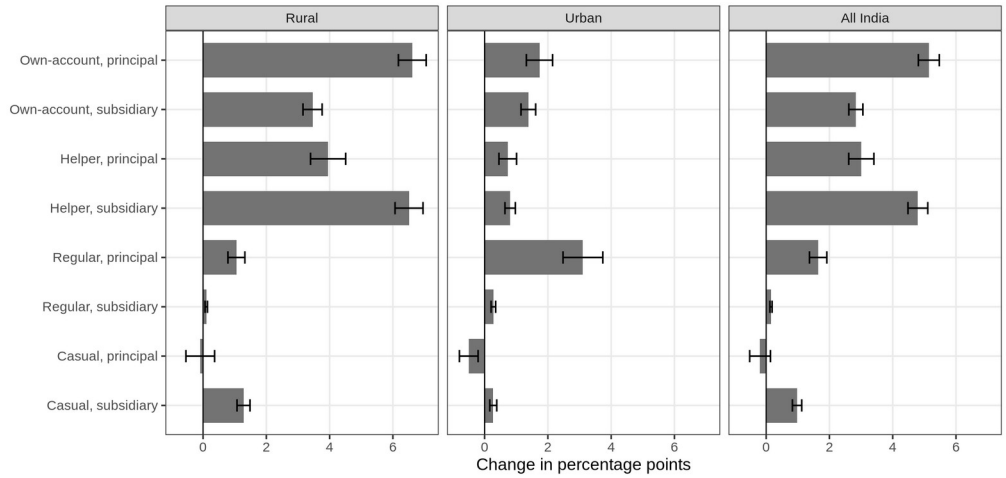


Fig. 1 *Contributions to the change in women's worker-population ratio, 2017–18 to 2023–24, by status and capacity (percentage points). Bars show point estimates; whiskers show 95 percent confidence intervals*

From domestic duties to work

The rise in work was mirrored by a fall in the share of women whose principal activity was domestic duties. Because the two surveys are independent cross-sections, these are net changes in shares, not observed transitions of individual women. Table 3 shows the distribution of women by usual principal activity. Among rural women, the share whose principal activity was domestic duties only (code 92) fell from 42.0 to 30.4 percent, while the share in domestic duties combined with free collection and similar activities for the household (code 93) was almost unchanged (18.4 and 18.0 percent). Principal-status unpaid helpers rose from 7.7 to 11.6 percent and principal-status own-account workers from 3.8 to 10.4 percent. Together, the principal-status work categories of rural women rose by 11.5 points, close to the fall in the domestic-duties-only share. The shares in education and in other categories outside the labor force changed little. In urban areas, too, the domestic-duties-only share fell, from 55.6 to 49.5 percent.

The rise in subsidiary-only work took place within the domestic-duties categories, since subsidiary-only workers are, by construction, women whose principal activity is outside the labor force. In 2023–24, 44.4 percent of rural subsidiary-only women workers had domestic duties only (code 92) as their principal status and 45.8 percent had code 93, proportions similar to 2017–18. What changed is the number of such women: the subsidiary-only WPR of rural women

rose from 2.8 to 14.1 percent. The rise in women's employment therefore consists to a large extent of women who are still mainly engaged in domestic duties but now report, or are recorded as having, a secondary economic activity.

The distinction between principal and subsidiary status matters for interpretation because the two statuses describe different relations to work. A principal worker spent most of the year in an economic activity. A subsidiary-only worker reports that her main activity was outside the labor force, usually domestic duties, and that she also did some economic work during the year, above a minimum threshold of days. The headline measure adds the two groups, but they are not equivalent. Taken at face value, the rise in subsidiary-only work is not a movement of women out of the home into a job. It is an increase in the number of women who combine domestic duties with some economic activity. Three readings are consistent with that pattern. Women may have started activities that they did not do before. Women who already did such work may have done more of it, so that it crossed the threshold. Or the same activity may have been recorded where it earlier went unrecorded. Each reading has different implications for hours, for earnings and for the reliability of the series, and the data separate them only partly. The porous boundary between domestic and economic activity is a long-standing theme in the literature on time use (Deshpande and Kabeer, 2024; Hirway and Jose, 2011). Panel evidence of frequent movement of Indian women into and out of work suggests that many subsidiary workers may be in transitional or intermittent positions (Sarkar, Sahoo, and Klasen, 2019).

Table 3 *Women aged 15 and over by usual principal activity status (percent)*

Usual principal activity status	Rural 2017–18	Rural 2023–24	Rural change	Urban 2017–18	Urban 2023–24	Urban change
Own-account worker or employer	3.8	10.4	6.6	3.8	5.5	1.7
Unpaid helper	7.7	11.6	4.0	1.7	2.4	0.7
Regular wage or salaried	2.5	3.5	1.1	9.4	12.5	3.1
Casual labor	7.0	6.9	-0.1	2.2	1.7	-0.5
Unemployed	1.0	1.2	0.3	2.2	2.1	-0.1
Attending education	10.4	10.9	0.5	13.0	12.1	-0.9
Domestic duties only (92)	42.0	30.4	-11.6	55.6	49.5	-6.1
Domestic duties and free collection etc. (93)	18.4	18.0	-0.3	4.7	6.9	2.2
Other, not in labor force	7.4	7.0	-0.4	7.4	7.1	-0.2

Note. Principal status only; women in codes 92, 93, education and other categories may also be subsidiary workers. Columns sum to 100.

Intensity of work

A worker in usual status may have worked very little in any given week. Table 4 reports, for women who are usual-status workers, the share who were employed in the reference week by current weekly status, mean hours actually worked in the week (including zero for those who did not work) and the share who worked fewer than 20 hours. Among all women workers, the share employed in the reference week fell from 86.2 to 83.7 percent, mean weekly hours fell from 36.6 to 28.2, and the share working fewer than 20 hours rose from 20.2 to 32.0 percent.

Part of the fall in average hours reflects the changed composition of workers. Subsidiary-only workers worked few hours in both years: subsidiary-only helpers averaged 12.6 hours in 2017–18 and 10.7 hours in 2023–24, and 76 percent of them worked fewer than 20 hours in 2023–24. Their number rose several-fold. Hours also fell within the principal-status groups, by 3.4 hours for principal-status helpers and 5.9 hours for principal own-account workers.

A summary measure that is not affected by where the line between workers and non-workers is drawn is the mean weekly hours of work per woman in the population. Because first visits are spread over all four quarters of the survey year, it averages over seasons, although a single reference week can still miss intermittent work. It rose from 8.12 (SE 0.094) to 11.39 (SE 0.094) hours, an increase of 40 percent. The WPR rose by 84 percent over the same period. In rural areas, hours per woman rose from 8.15 to 12.37, and in urban areas from 8.05 to 9.13. One caution is that weekly hours per worker also fell for men, from 50.6 to 44.9, and hours per man in the population fell from 36.04 to 34.24. Part of the fall in hours per worker may therefore reflect changes in how hours were reported rather than changes specific to women; if so, the increase in hours per woman may be understated. Even so, the rise in women’s hours of work is about half the rise in the share of women counted as workers.

Table 4 *Intensity of work among usual-status workers aged 15 and over*

	Employed in reference week (%) 2017–18	Employed in reference week (%) 2023–24	Mean hours worked in week 2017–18	Mean hours worked in week 2023–24	Worked under 20 hours (%) 2017–18	Worked under 20 hours (%) 2023–24
All women workers	86.2	83.7	36.6	28.2	20.2	32.0
Rural	83.7	81.8	34.1	26.6	23.2	34.9
Urban	93.5	91.6	44.0	35.1	11.4	20.0
Own-account, principal	95.0	96.5	40.1	34.2	10.8	15.5
Own-account, subsidiary only	69.7	82.3	17.7	17.2	51.5	55.7

	Employed in reference week (%) 2017–18	Employed in reference week (%) 2023–24	Mean hours worked in week 2017–18	Mean hours worked in week 2023–24	Worked under 20 hours (%) 2017–18	Worked under 20 hours (%) 2023–24
Unpaid helper, principal	85.2	89.4	34.3	30.9	20.4	23.7
Unpaid helper, subsidiary only	48.8	54.4	12.6	10.7	68.6	76.4
Regular wage, principal	98.5	98.3	50.7	42.1	2.4	6.4
Casual labor, principal	84.0	83.0	34.1	29.2	23.4	26.4
Casual labor, subsidiary only	41.7	32.0	11.8	9.2	71.9	76.3
All men workers (comparison)	96.3	96.7	50.6	44.9	5.8	7.7

Note. Hours actually worked in all economic activities in the seven days before the survey, including zero for workers who did not work in that week. Regular wage workers in a subsidiary capacity only are few and not shown.

Production for the household's own use

The 2023–24 survey asked self-employed persons in primary-sector activities (NIC divisions 01–09) about the destination of their output. Among women own-account workers and helpers in these activities, 16.2 percent reported that the output was for own consumption only, 25.9 percent that it was mainly for own consumption with some sale, 51.8 percent that it was mainly for sale and 6.1 percent that it was entirely for sale (Table 5). The corresponding share for own consumption only or mainly was 25.5 percent among men. The question was answered for all of these workers. Production mainly for own use was most common among rural women working in a subsidiary capacity: 52.3 percent of them produced only or mainly for the household, compared with 36.3 percent of rural women working in a principal capacity.

Under the 2013 international definition, production of goods intended mainly for the household's own final use is own-use production work, not employment (International Labour Organization, 2013). Table 5 shows how the 2023–24 WPR would change if such work were excluded. Excluding women whose output was for own consumption only lowers the women's WPR from 40.3 to 37.0 percent; also excluding those whose output was mainly for own consumption lowers it to 31.5 percent. For rural women, the corresponding values are 46.5, 41.9 and 34.3 percent. These are approximations: the classification is by destination of output and applies only to self-employment in primary activities, and some excluded women may have other work that the survey does not record. They also cannot be compared with 2017–18, when the question was not asked. They show, however, that a substantial part of measured employment of women in 2023–

24 is production mainly for the household, which the 2013 standard classifies as own-use production work rather than employment.

Table 5 *Destination of output of self-employed workers in primary activities, and the worker-population ratio excluding production for own use, 2023–24 (percent)*

Panel A: destination of output	Women, rural, principal	Women, rural, subsidiary only	Women, all	Men, all
Own consumption only	12.8	21.8	16.2	5.1
Mainly own consumption, some sale	23.4	30.5	25.9	20.4
Mainly for sale	56.8	43.3	51.8	66.0
Entirely for sale	6.9	4.4	6.1	8.5
Panel B: WPR, 2023–24	*Women, rural*	*Women, urban*	*Women, all*	*Men, all*
WPR as published definition	46.5	26.0	40.3	76.3
Excluding own consumption only	41.9	25.6	37.0	75.1
Excluding own consumption only or mainly	34.3	25.1	31.5	70.4

Note. Column headings for Panel B are given in its first row. Panel A: own-account workers, employers and unpaid helpers in NIC 2008 divisions 01–09, by capacity in which the activity is pursued; columns sum to 100. Panel B: worker-population ratio of persons aged 15 and over, after excluding self-employed workers in these divisions whose output was for own consumption only, or only or mainly. The question was not asked in 2017–18.

Composition

Women's employment varies strongly with age, education, marital status and place of residence, so part of the change in the WPR could reflect changes in the composition of the population between the two surveys. Table 6 shows that it does not. With 4,093 cells defined by state, sector, age group, education and marital status, the composition component is -1.06 points and the rate component is 19.42 points. Composition slightly lowered the expected ratios of casual labor and unpaid help and slightly raised that of regular wage employment. The composition component is between -1.06 and 0.06 points for all sets of cells considered, including cells that add social group (Table A2, Panel A). Restricting the population to women of working age (15–59) gives the same picture: their WPR rose by 19.9 points, of which 9.4 points were subsidiary-only work (Table A2, Panel B). The rise in women's employment took place within groups of women, not through a shift toward groups with higher employment.

Table 6 *Kitagawa decomposition of the change in women's worker-population ratio, 2017–18 to 2023–24 (percentage points)*

	Total change	Composition	Rates within cells
Worker-population ratio	18.36	-1.06	19.42
Own-account worker or employer	7.98	0.01	7.97
Unpaid helper	7.80	-0.55	8.35
Regular wage or salaried	1.80	0.41	1.39
Casual labor	0.79	-0.92	1.71

Note. Cells: state by sector by age group by education by marital status (4,093 cells present in both years). Women aged 15 and over.

Who and where

The rise was broad-based but uneven (Table 7). Among rural women, the WPR rose by 13.7 points for those aged 15–24 and by 31.6 and 32.6 points for those aged 35–44 and 45–59. It rose more for women with below-primary (26.0 points) or primary or middle schooling (27.5 points) than for graduates (12.1 points). It rose in every quintile of household consumption per person, least in the poorest (17.9 points) and by 25.3 and 23.1 points in the middle and richest quintiles. It rose most among Scheduled Tribe women, from 37.4 to 66.9 percent. Among the youngest rural women, the rise came mostly through subsidiary work (9.5 of 13.7 points). In urban areas the increases were smaller in every group.

Table 7 *Women's worker-population ratio by group, women aged 15 and over (percent)*

	Rural 2017–18	Rural 2023–24	Rural change	Urban 2017–18	Urban 2023–24	Urban change
Age group						
15–24	9.3	23.0	13.7	8.4	13.8	5.3
25–34	27.2	48.0	20.8	22.9	30.6	7.7
35–44	36.0	67.5	31.6	27.6	38.7	11.1
45–59	33.9	66.5	32.6	22.1	32.4	10.4
60+	11.7	27.1	15.4	7.4	10.2	2.7
Education						
Below primary	29.1	55.2	26.0	22.0	30.3	8.3
Primary or middle	20.8	48.3	27.5	16.2	26.7	10.5
Secondary or higher secondary	14.8	32.4	17.6	11.1	17.2	6.1
Graduate or above	20.9	33.0	12.1	26.7	32.5	5.9
Household consumption quintile						
Q1 (poorest)	24.8	42.7	17.9	17.2	24.4	7.3
Q2	22.8	45.3	22.4	17.7	25.9	8.2
Q3	23.1	48.3	25.3	18.2	25.6	7.4
Q4	24.2	49.3	25.0	18.1	26.3	8.2
Q5 (richest)	23.4	46.5	23.1	19.5	27.4	8.0
Social group						
Scheduled Tribe	37.4	66.9	29.5	22.4	36.3	13.9
Scheduled Caste	24.2	46.2	22.0	22.9	30.0	7.2
Other Backward Class	22.8	45.8	23.0	18.5	25.4	6.8

	Rural 2017–18	Rural 2023–24	Rural change	Urban 2017–18	Urban 2023–24	Urban change
Others	18.3	36.7	18.4	15.8	24.1	8.2

Note. Usual status (ps+ss). Consumption quintiles are of monthly household consumption expenditure per person, within year and sector.

Across the 20 most populous states, the change in women's WPR ranged from 2.3 points in Andhra Pradesh and 5.5 points in Delhi to 35.0 points in Jharkhand and 37.2 points in Assam (Fig. 2; Table A3 gives the state values). Across these states, the correlation between the change in the WPR and the change in the combined ratio of own-account workers and helpers is 0.97. States differed in which of the two categories rose: in Jharkhand, Odisha and Madhya Pradesh it was mainly unpaid helpers, in Assam and West Bengal mainly own-account workers. Because self-employment accounts for most of the national rise, this correlation is partly mechanical. The exceptions are states with smaller rises in which wage work accounted for half or more of the change, such as Kerala, Tamil Nadu and Delhi (Table A3). Elsewhere the pattern of a rise concentrated in self-employment holds, so the national result does not reflect a few states with unusual estimates.

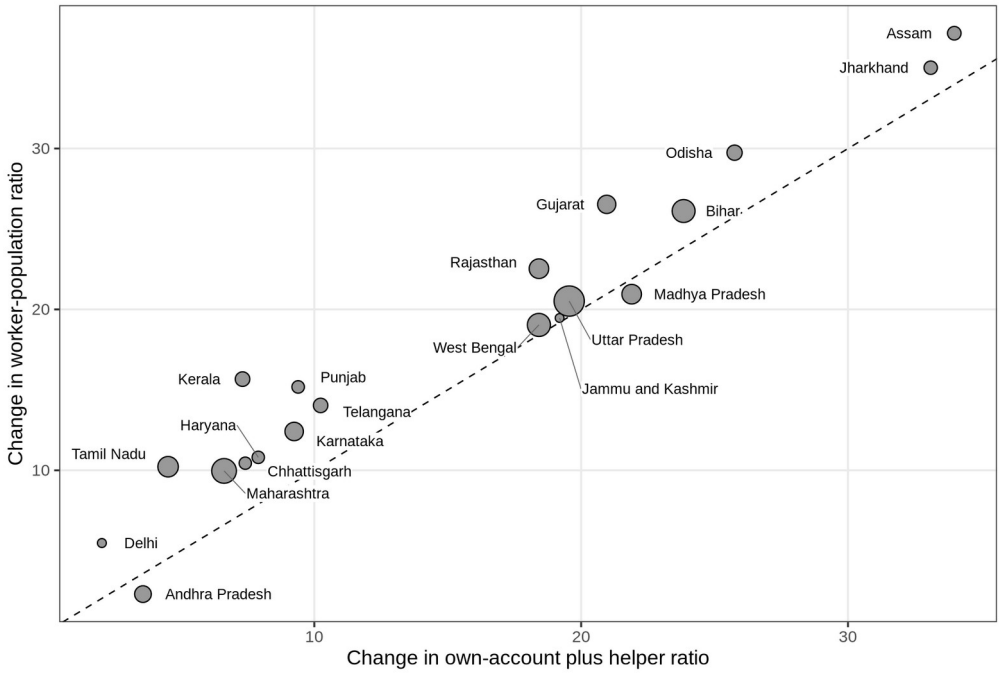


Fig. 2 *Change in women's worker-population ratio and in the self-employed (own-account plus helper) ratio, 20 most populous states, 2017–18 to 2023–24 (percentage points). Marker area is proportional to population; the dashed line marks equal changes*

Discussion and Conclusion

The accounting establishes four facts about the measured rise in Indian women's employment between 2017–18 and 2023–24. It was a rise in self-employment, mostly in agriculture and mostly in rural areas. Nearly half of it consisted of women whose main activity remained domestic duties and who were counted as workers through a subsidiary activity. It amounted to a much smaller increase in hours of work than in the number of workers. And a large share of the self-employment of women in primary production in 2023–24 was production mainly for the household's own consumption.

Competing interpretations

These facts are consistent with more than one interpretation, and the data cannot separate them. If more women took up or expanded work on family farms and livestock because of pressure on household incomes, the rise would look like this (Azim Premji University, 2023; Jha, Ghatak, and Singh, 2026). If additional family labor reflects productive opportunities in agriculture and

allied activities, the pattern would also look like this (Goldar and Aggarwal, 2026). And if the same activities were recorded more completely, after field instructions that direct enumerators to probe for kitchen gardens, poultry, dairy and free collection and explain when such activity makes a woman engaged in domestic duties a subsidiary worker, the rise would again look like this (Thakur and Chaudhary, 2025). The categories that rose are the ones most sensitive to probing in the measurement literature (Bardasi et al., 2011; Gaddis et al., 2021; Langsten and Salen, 2008): unpaid family work, subsidiary activity, small-scale livestock and collection, and production for own use.

Some patterns bear on the relative importance of these explanations without settling it. The worked examples in the instructions describe women who would otherwise be coded 93 being recorded instead with principal status 92 and a subsidiary activity; such a reclassification would, other things equal, lower the share in code 93. That share hardly changed among rural women and rose among urban women, while the share in code 92 fell, mainly because more women reported a principal economic activity. The rise was also gradual: in the published series, the women's WPR was 22.0, 23.3, 28.7, 31.4, 31.7, 35.9 and 40.3 percent in the seven years from 2017–18 to 2023–24 (National Sample Survey Office, 2024), so about half of the rise had occurred by 2021–22, before the worked examples that the survey's note dates to 2022–23. An explanation that rests only on the 2023–24 instructions therefore cannot account for the whole rise, although the undated new item in the list of activities needing special care may have been introduced earlier. These patterns are suggestive rather than diagnostic. And men also show a rise in unpaid help and subsidiary work, on a smaller scale. A full assessment would need the annual unit-level data for all years and, ideally, an experimental comparison of probing versions within the survey.

Several threats to comparability across the two rounds deserve explicit mention, apart from the change in instructions. The first is seasonality and the reference period. The usual-status measure covers a full year, so it depends on recall, and the weekly measure depends on the week of the interview. Both can respond differently to changes in the share of the year that women spend in agricultural tasks. The second is the identity of the respondent. Survey experiments show that whether a woman answers for herself or another household member answers for her affects measured employment (Bardasi et al., 2011). A change in the practice of who is interviewed

would therefore change the series without any change in behavior. The third is the quality of enumeration and supervision. Experimental evidence shows that the design of recall questions changes recorded farm labor (Gaddis et al., 2021), and the field note on updated instructions shows that guidance evolved between rounds (National Sample Survey Office, 2023b). The fourth is the sampling frame and the design weights; if the frame represents some groups of households better in one round than in the other, levels can shift without any change in within-group rates. The fifth is the economic setting. The period includes a pandemic and its aftermath, and household enterprises and agricultural work may have absorbed labor for reasons unrelated to survey procedure. None of these threats can be assessed with two cross-sections alone, and none should be treated as established. They indicate what a fuller test of comparability would need.

What the headline rate does and does not mean for policy

Whatever the mix of causes, the findings have implications for how the rise is used. Measured as hours, the increase in women's work is about half as large as measured by the WPR. Measured against the 2013 international definition of employment, the women's WPR in 2023–24 would be between 31.5 and 37.0 percent rather than 40.3 percent. Comparisons of India's women's employment with other countries, or with earlier Indian estimates, should therefore use comparable measures: hours, principal status, or employment excluding own-use production.

The headline rate is often used as a summary of women's economic participation, in national assessments and in international comparisons. The results suggest what it does and does not tell users. It does tell them that many more women are recorded as doing some economic activity in the year, and that this is true within groups of women defined by age, education, marital status, sector and state. It does not by itself tell them that women's paid employment, earnings or hours of market work rose in proportion. Regular wage employment accounts for 10 percent of the rise; unpaid help, which by definition brings no direct individual income, accounts for 42 percent; and a large part of the self-employment of women in primary production is mainly for the household's own consumption. Programs whose objectives concern women's earnings, social-security coverage or access to wage work are better monitored with measures of paid employment and earnings than with the headline rate. Programs concerned with the time burden on women, such as those providing water, fuel or childcare, are better monitored with hours of

work and time use, because the activities that rose (collection, kitchen gardens, livestock) compete with domestic work for women's time. And targets expressed as a women's participation or employment rate are best accompanied by the principal-status rate and an hours-based measure, so that progress toward them can be read correctly.

The PLFS already contains the information needed to produce such measures. Publishing them alongside the headline rates, and reporting principal and subsidiary workers separately, would make the series easier to interpret without new data collection. The same applies to the destination-of-output question: tabulating the employment rate with and without production mainly for own use would let users compare India's figures with those of countries that have adopted the 2013 standards.

The same point applies to international gender monitoring. Global frameworks, including the Sustainable Development Goal indicators on gender equality and decent work, rely on a small set of measures that national statistical offices report: employment and participation rates, the share of employment that is informal, the time spent on unpaid domestic and care work, and wage or earnings gaps. A rise in a women's employment rate is normally read in these frameworks as progress. The Indian case shows that the rate alone can be a poor guide when the rise lies in unpaid or own-use forms of work. Indicators that count the same activity both as employment and as unpaid work would send contradictory signals, and cross-country rankings would depend on whether a country has adopted the 2013 definitions (International Labour Organization, 2013). A practical response is to report, for each country, a core set of linked indicators: the employment rate under the 2013 definition, the rate under the broader national concept, the principal-status rate, and hours. Countries that keep the broader concept, as India does, could report the destination-of-output tabulation as a bridge. Such reporting does not judge any national choice. It allows users to see which concept underlies a figure before they compare it, and it reduces the risk that a measurement change is read as a change in women's economic position.

Lessons for surveys elsewhere

The Indian case also carries lessons for other countries whose labor force surveys are moving to the 2013 standards. First, where a large share of women's work is in agriculture, livestock and

collection, the measured employment rate of women can change substantially with changes in probing, even when definitions are fixed; survey experiments have shown this within a single survey (Bardasi et al., 2011; Benes and Walsh, 2018; Gaddis et al., 2021), and the PLFS shows it is compatible with a large movement in a national series. Second, separating principal from subsidiary activity, and persons from hours, provides an internal check on whether a rise in the employment rate reflects more work or more workers. Third, a single question on the destination of output allows a survey to report both the older, broader concept of work and an employment rate consistent with the new standards, which eases the transition between concepts. Fourth, changes to field instructions are part of the measurement system; dating and publishing them, and where possible testing them experimentally, would make it easier to separate changes in women's work from changes in how it is recorded.

Limitations and conclusion

The analysis has limitations. It compares two years six years apart, and the period includes the COVID-19 pandemic; the end points do not show the path. The decomposition is an accounting, not a causal analysis. The destination-of-output question is available only for 2023–24, so the own-use adjustment cannot be applied to 2017–18. The fall in reported hours for men suggests that hours may not be fully comparable between the years. And the assessment of changes in field instructions is based on the published instructions; the data cannot show how they were applied.

Several kinds of additional data would do more than a third cross-section to settle what the rise means. The first is the annual unit-level data for every year in the series, which would allow the timing of the rise to be compared with the timing of each change in instructions, and would show whether jumps coincide with documented changes. The second is a split-sample experiment inside the survey, in which a random part of the sample is interviewed with the older instructions and the rest with the current ones. Experiments of this kind have been used elsewhere to measure the effect of questionnaire design on women's recorded work (Bardasi et al., 2011; Gaddis et al., 2021), and they are the cleanest way to separate recording from behavior. The third is longitudinal information on the same women, which would show whether subsidiary workers remain in that status, move to principal work or return to domestic duties. The fourth is linked time-use data, which would show how the added work is distributed over the day and how it

relates to domestic work (Hirway and Jose, 2011). The fifth is plot-level or enterprise-level information on household farms, which has been valuable in other settings for understanding who supplies labor to which tasks (Palacios-Lopez, Christiaensen, and Kilic, 2017). Process information on interviews, such as the respondent and the interview length, would also help. Together, these sources would let future work move from an accounting of the rise to a test of its causes (Thakur and Chaudhary, 2025).

The rise in women's employment in India between 2017–18 and 2023–24 is reproduced, to within rounding, by the unit-level data, and it is real in the sense that the survey records many more women as workers under formally unchanged definitions, although the guidance to enumerators changed. It is concentrated in own-account work and unpaid help in agriculture, in rural areas, among women who work in a subsidiary capacity alongside domestic duties, and in production often mainly for the household's own use. It is not explained by changes in the composition of the population, and in hours of work it is about half as large. The rise therefore lies largely in forms of work close to the boundary of what surveys count as work. Whether it reflects new work, more work by women already working, or more complete recording of existing work matters for how it is read, and answering that question will require the measures this paper proposes to be tracked over time, in India and in other countries where much of women's work lies near that boundary.

Statements and Declarations

Funding. No funding was received for conducting this study.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Ethics approval. The study is a secondary analysis of anonymized official survey data released for research use. It involved no contact with human participants, and no ethics approval was required.

Data availability. The unit-level data of the Periodic Labour Force Survey 2017–18 and 2023–24 are produced by the National Statistics Office, Ministry of Statistics and Programme Implementation, Government of India, and are available from the Ministry's microdata portal

under its terms of use. The analysis code is available from the author on request and will be deposited in a public repository on acceptance.

Use of AI tools. The author used Claude (Anthropic) to improve the language and readability of parts of the text and to assist with code review and checking. The author reviewed and verified all outputs and takes full responsibility for the content.

References

Afridi, F., Dinkelman, T., and Mahajan, K. (2018). Why are fewer married women joining the work force in rural India? A decomposition analysis over two decades. *Journal of Population Economics*, *31*(3), 783–818. https://doi.org/10.1007/s00148-017-0671-y

Andres, L. A., Dasgupta, B., Joseph, G., Abraham, V., and Correia, M. (2017). *Precarious drop: Reassessing patterns of female labor force participation in India* (Policy Research Working Paper No. 8024). World Bank. https://doi.org/10.1596/1813-9450-8024

Ara, S., and Shrivastav, P. K. (2025). Decoding rising female labour force participation in India: Evidence from PLFS data. *Economic and Political Weekly*, *60*(26–27). https://doi.org/10.71279/epw.v60i26-27.39982

Azim Premji University. (2023). *State of working india 2023: Social identities and labour market outcomes*. Centre for Sustainable Employment, Azim Premji University. https://azimpremjiuniversity.edu.in/publications/2023/report/state-of-working-india-2023-social-identities-and-labour-market-outcomes

Bardasi, E., Beegle, K., Dillon, A., and Serneels, P. (2011). Do labor statistics depend on how and to whom the questions are asked? Results from a survey experiment in Tanzania. *The World Bank Economic Review*, *25*(3), 418–447. https://doi.org/10.1093/wber/lhr022

Beneria, L. (1981). Conceptualizing the labor force: The underestimation of women's economic activities. *The Journal of Development Studies, 17*(3), 10–28. https://doi.org/10.1080/00220388108421795

Benes, E. M., and Walsh, K. (2018). *Measuring employment in labour force surveys: Main findings from the ILO LFS pilot studies* (Statistical Methodology Series No. 4). International Labour Organization, Department of Statistics. https://www.ilo.org/publications/report-iv-measuring-employment-labour-force-surveys-main-findings-ilo-lfs

Chatterjee, E., Desai, S., and Vanneman, R. (2018). Indian paradox: Rising education, declining women's employment. *Demographic Research, 38*, 855–878. https://doi.org/10.4054/DemRes.2018.38.31

Desai, S., and Joshi, O. (2019). The paradox of declining female work participation in an era of economic growth. *The Indian Journal of Labour Economics, 62*(1), 55–71. https://doi.org/10.1007/s41027-019-00162-z

Deshpande, A., and Kabeer, N. (2024). Norms that matter: Exploring the distribution of women's work between income generation, expenditure-saving and unpaid domestic responsibilities in India. *World Development, 174*, 106435. https://doi.org/10.1016/j.worlddev.2023.106435

Gaddis, I., and Klasen, S. (2014). Economic development, structural change, and women's labor force participation: A reexamination of the feminization U hypothesis. *Journal of Population Economics*, *27*(3), 639–681. https://doi.org/10.1007/s00148-013-0488-2

Gaddis, I., Oseni, G., Palacios-Lopez, A., and Pieters, J. (2021). Measuring farm labor: Survey experimental evidence from Ghana. *The World Bank Economic Review*, *35*(3), 604–634. https://doi.org/10.1093/wber/lhaa012

Goldar, B., and Aggarwal, S. C. (2026). Is the productivity of unpaid female family workers in Indian agriculture very low? *Economic and Political Weekly*, *61*(5). https://doi.org/10.71279/epw.v61i5.46716

Goldin, C. (1994). *The U-shaped female labor force function in economic development and economic history* (NBER Working Paper No. 4707). National Bureau of Economic Research. https://doi.org/10.3386/w4707

Hirway, I., and Jose, S. (2011). Understanding women's work using time-use statistics: The case of India. *Feminist Economics*, *17*(4), 67–92. https://doi.org/10.1080/13545701.2011.622289

International Labour Organization. (2013). *Resolution concerning statistics of work, employment and labour underutilization, adopted by the nineteenth international conference of labour statisticians*. Geneva: ILO. https://www.ilo.org/resource/resolution-concerning-statistics-work-employment-and-labour-0

Jayachandran, S. (2021). Social norms as a barrier to women's employment in developing countries. *IMF Economic Review*, *69*(3), 576–595. https://doi.org/10.1057/s41308-021-00140-w

Jha, M., Ghatak, M., and Singh, J. (2026). Rise of unpaid family workers: Evidence on distress-driven employment growth in India. *Economic and Political Weekly, 61*(1). https://doi.org/10.71279/epw.v61i01.46999

Kitagawa, E. M. (1955). Components of a difference between two rates. *Journal of the American Statistical Association*, *50*(272), 1168–1194. https://doi.org/10.1080/01621459.1955.10501299

Klasen, S. (2019). What explains uneven female labor force participation levels and trends in developing countries? *The World Bank Research Observer*, *34*(2), 161–197. https://doi.org/10.1093/wbro/lkz005

Klasen, S., and Pieters, J. (2015). What explains the stagnation of female labor force participation in urban India? *The World Bank Economic Review*, *29*(3), 449–478. https://doi.org/10.1093/wber/lhv003

Lahoti, R., and Swaminathan, H. (2016). Economic development and women's labor force participation in India. *Feminist Economics*, *22*(2), 168–195. https://doi.org/10.1080/13545701.2015.1066022

Langsten, R., and Salen, R. (2008). Two approaches to measuring women's work in developing countries: A comparison of survey data from Egypt. *Population and Development Review*, *34*(2), 283–305. https://doi.org/10.1111/j.1728-4457.2008.00220.x

Lumley, T. (2004). Analysis of complex survey samples. *Journal of Statistical Software, 9*(8). https://doi.org/10.18637/jss.v009.i08

Mehrotra, S., and Parida, J. K. (2017). Why is the labour force participation of women declining in India? *World Development*, *98*, 360–380. https://doi.org/10.1016/j.worlddev.2017.05.003

National Sample Survey Office. (2016). *Periodic labour force survey: Instructions to field staff, volume I: Design, concepts, definitions and procedures*. New Delhi: Ministry of Statistics and Programme Implementation, Government of India. Released with the unit-level data for July 2017 – June 2018.

National Sample Survey Office. (2023a). *Periodic labour force survey: Instructions to field staff, volume I: Design, concepts, definitions and procedures*. New Delhi: Ministry of Statistics and Programme Implementation, Government of India. Released with the unit-level data for July 2023 – June 2024.

National Sample Survey Office. (2023b). *Updated instruction to field staff of periodic labour force survey (PLFS) for July 2023 – June 2024*. New Delhi: Ministry of Statistics and Programme Implementation, Government of India. Released with the unit-level data.

National Sample Survey Office. (2024). *Press note on periodic labour force survey (PLFS) annual report [July 2023 – June 2024]*. New Delhi: Ministry of Statistics and Programme Implementation, Government of India, 23 September. https://www.mospi.gov.in/sites/default/files/press_release/Press_note_AR_PLFS_2023_24_22092024.pdf

Palacios-Lopez, A., Christiaensen, L., and Kilic, T. (2017). How much of the labor in African agriculture is provided by women? *Food Policy*, *67*, 52–63. https://doi.org/10.1016/j.foodpol.2016.09.017

Sarkar, S., Sahoo, S., and Klasen, S. (2019). Employment transitions of women in India: A panel analysis. *World Development, 115*, 291–309. https://doi.org/10.1016/j.worlddev.2018.12.003

Thakur, A. N., and Chaudhary, P. (2025). Decoding the recent rise in women's work participation. *The Indian Journal of Labour Economics*, *68*(4), 1455–1474. https://doi.org/10.1007/s41027-025-00601-0

Appendix

Table A1 *Comparison of field instructions relevant to measuring women's work, PLFS 2017–18 and 2023–24*

Element	2017–18	2023–24
Definition of economic activity: primary production for own consumption, including free collection	Included	Included
Codes 92 (domestic duties only) and 93 (domestic duties and free collection etc. for household use)	Included	Included
Subsidiary status: economic activity for 30 days or more in the reference year	Included	Included
List of activities needing special care: kitchen garden, orchards, household poultry, dairy, free collection of fish, small game, wild fruits, vegetables for household consumption or sale	Not included (list has items a–h)	Added as item (j) of Box 8
Worked examples distinguishing codes 92/93 from subsidiary self-employment (usual status)	Not included	Added (Box 9)
Worked examples for codes 92/93 in the daily record (current weekly status)	Not included	Added (Box 10)
Question on destination of output of self-employed in NIC 01–09	Not asked	Asked

Note. Source: Instructions to Field Staff, Volume I, for 2017–18 and 2023–24 (Chapter Three), and the note on updated instructions for 2023–24, which dates the clarifications in Boxes 9 and 10 to PLFS 2022–23.

Table A2 *Robustness: alternative decomposition cells and working ages*

Panel A: Kitagawa decomposition, alternative cells	Cells	Unmatched (%)	Composition	Rates
Cells: sector	2	0.00	0.06	18.30
Cells: state × sector	70	0.00	-0.06	18.42
Cells: sector × age × education	40	0.00	-0.69	19.06
Cells: state × sector × age × education × marital status	4,093	0.32	-1.06	19.42
Cells: state × sector × age × education × marital status × social group	10,625	1.26	-0.41	18.78

Panel A: Kitagawa decomposition, alternative cells	Cells	Unmatched (%)	Composition	Rates
Panel B: women aged 15–59	*2017–18*	*2023–24*	*Change*	*SE*
Worker-population ratio	23.8	43.7	19.9	0.38
Own-account	4.7	13.3	8.7	0.22
Unpaid helper	7.5	15.9	8.3	0.27
Regular wage	5.2	7.3	2.2	0.16
Casual	6.4	7.1	0.7	0.20
Subsidiary only	2.4	11.8	9.4	0.24

Note. Column headings for Panel B are given in its first row. Panel A: change in women's WPR (all India, 15+, percentage points) decomposed with alternative cell definitions; unmatched is the population share in cells present in one year only. Panel B: usual status (ps+ss), percent of women aged 15–59; change and its standard error in percentage points.

Table A3 *Change in women's worker-population ratio and its components, 20 most populous states, 2017–18 to 2023–24 (percentage points)*

State	Worker-population ratio	Own-account	Unpaid helper	Wage (regular and casual)
Assam	37.2	22.2	11.8	3.2
Jharkhand	35.0	12.0	21.1	1.9
Odisha	29.7	10.3	15.5	4.0
Gujarat	26.5	10.4	10.6	5.6
Bihar	26.1	10.6	13.3	2.3
Rajasthan	22.5	7.7	10.7	4.1
Madhya Pradesh	20.9	7.9	14.0	-0.9
Uttar Pradesh	20.5	9.8	9.7	1.0
Jammu and Kashmir	19.5	23.2	-4.0	0.3
West Bengal	19.0	12.3	6.1	0.6
Kerala	15.7	5.2	2.2	8.4
Punjab	15.2	7.3	2.1	5.8
Telangana	14.0	3.8	6.5	3.8
Karnataka	12.4	3.3	5.9	3.2
Haryana	10.8	3.6	4.3	2.9
Chhattisgarh	10.4	0.6	6.8	3.0
Tamil Nadu	10.2	4.1	0.4	5.7
Maharashtra	10.0	3.9	2.7	3.3
Delhi	5.5	0.8	1.2	3.4
Andhra Pradesh	2.3	0.7	2.9	-1.3

Note. Usual status (ps+ss), women aged 15 and over. States ordered by the change in the worker-population ratio; the three components sum to it. Jammu and Kashmir includes Ladakh.